1. (30 points) A company charges a fixed fee plus a constant amount per hour for consulting. When a client uses 2 hours of consulting, the total bill is $140. When a client uses 5 hours, the total bill is $260.

a) Find a linear function C(t) representing the total cost C in dollars as a function of time t in hours.

b) Interpret the slope and the y-intercept in the context of this problem.

c) Use your model to find the cost of 8 hours of consulting.

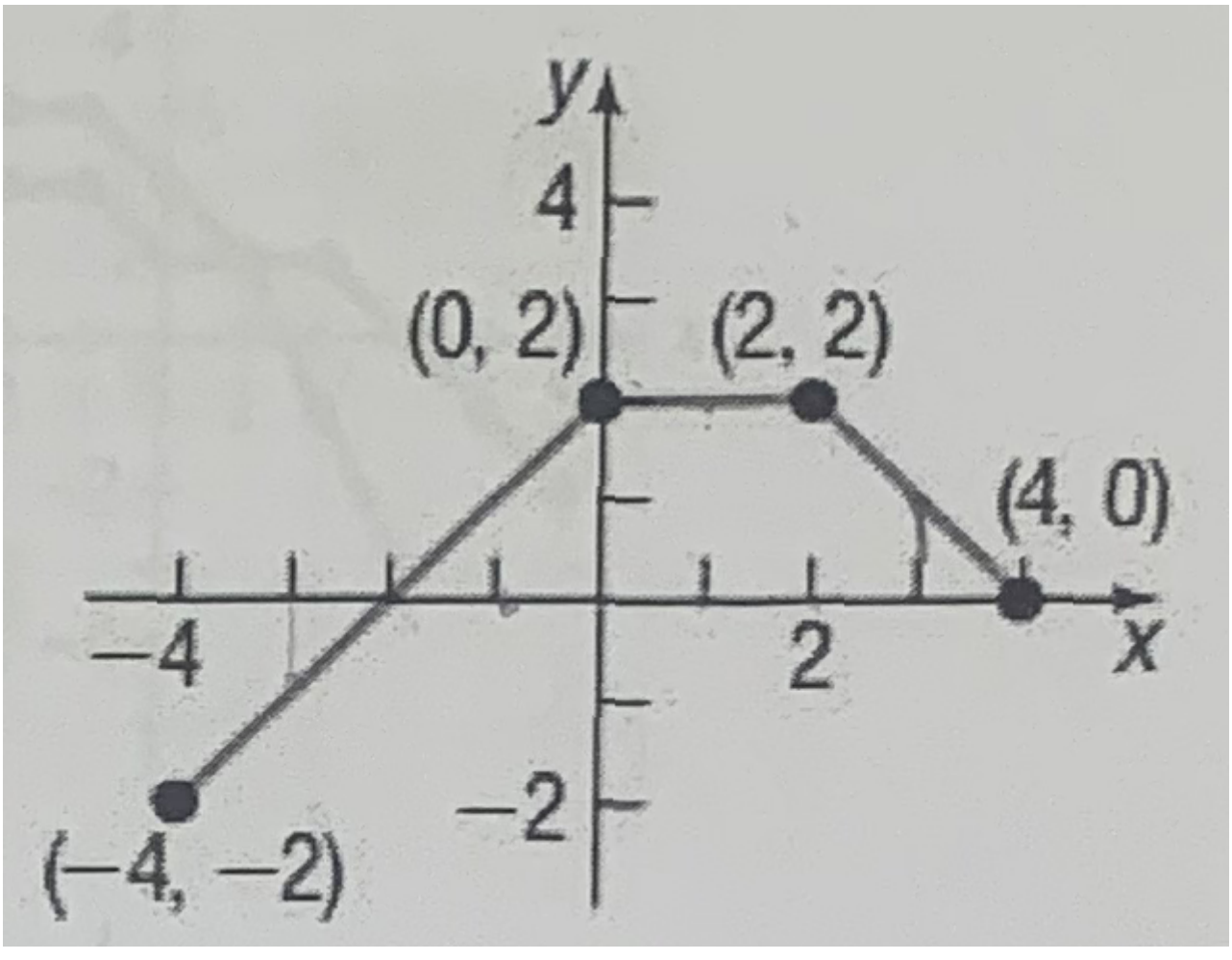
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2. (30 points) Below is the graph of the function y=f(x).

Draw the graph of the following functions:

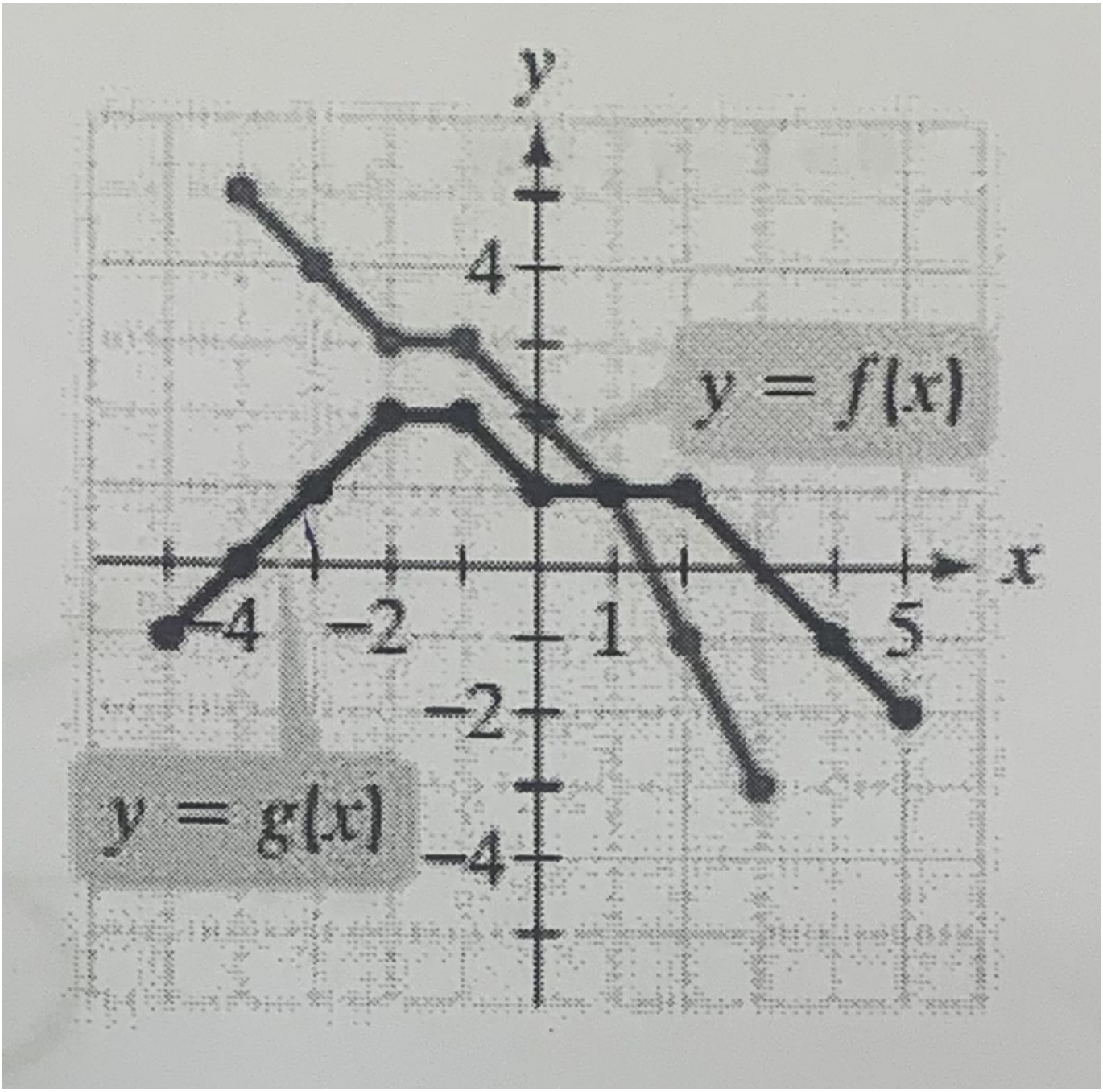
a. $y = f(x+1) - 2$

b. $y = -f(x)/2$



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3. (30 points) Use the graph of $f(x)$ and $g(x)$ in the picture below to find



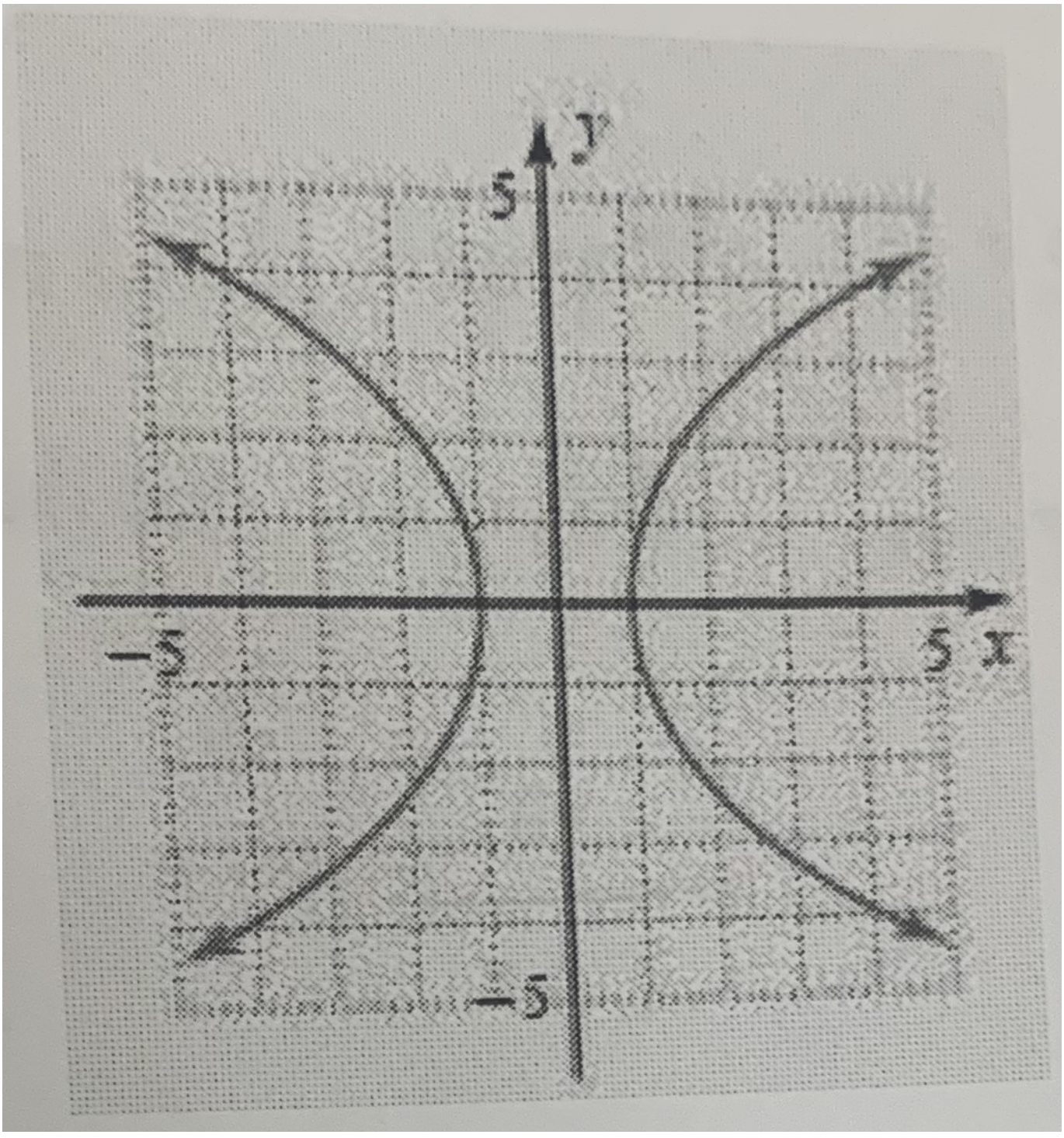
a. $(g-f)(-2) =$
b. $(fg)(-1) =$
c. $(f \circ g)(4) =$
d. $(g \circ f)(-2) =$
e. $(g/f)(2) =$

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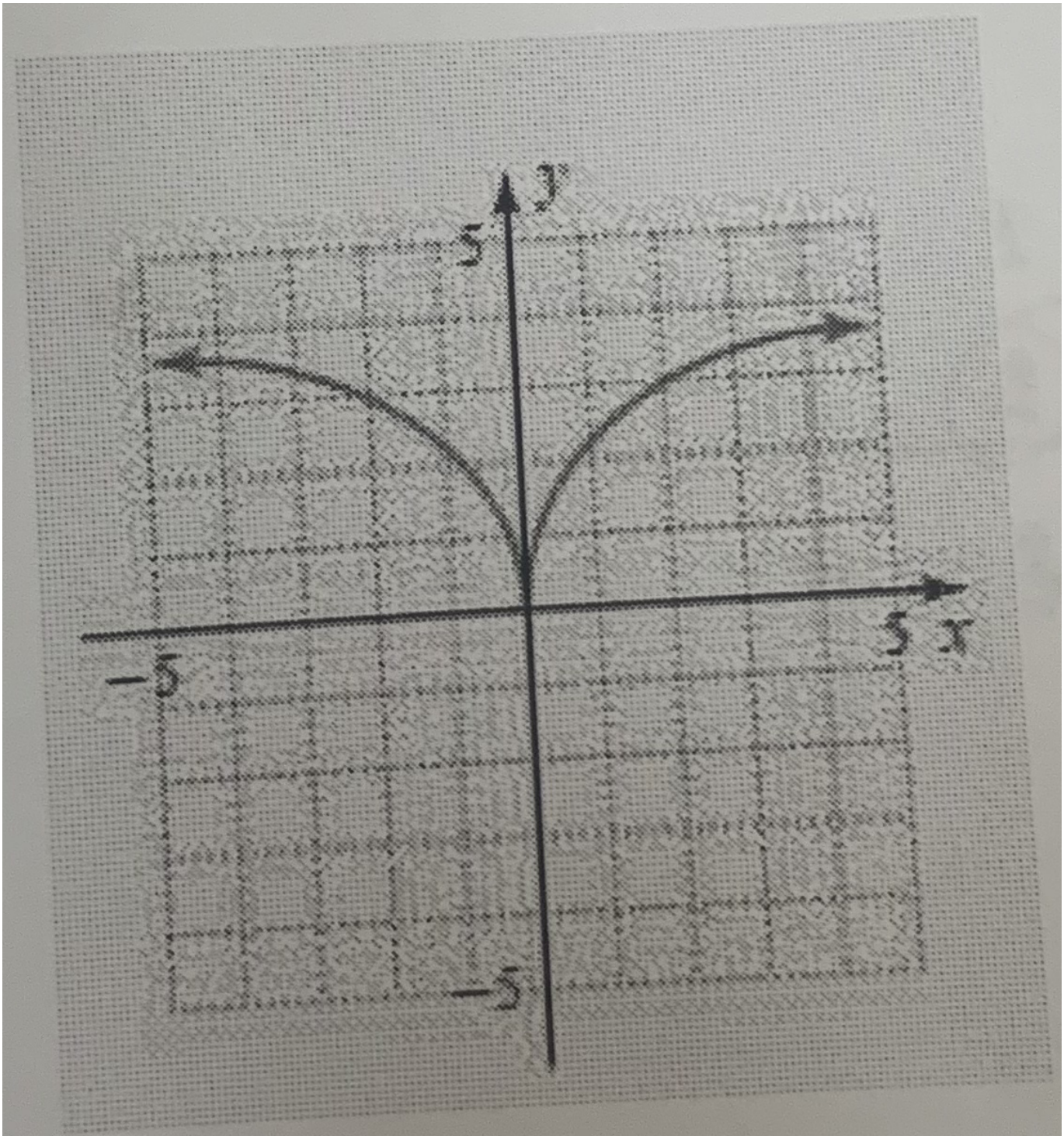
4. (30 points) Determine if the following equations and graphs represents y as a function of x? Explain your answer.

a. $x = |y-2|$

b.



c.



5. (40 points)

a) Find the asymptotes of the following functions $\frac{x^2+x-6}{x-3}$

b) Use the synthetic division to divide: $\frac{x^5+x^3-2}{x-1}$

6. (30 points)

You want to have $10,000 in an investment account at a future time. The account earns interest continuously at an annual rate of 5.6%. You plan to deposit an unknown amount P dollars now (at t=0).
After 3 years, you deposit an additional $2,000 into the same account.
After a total of 8 years from your initial deposit, the account balance is exactly $10,000.

(a) Write an expression for the amount in the account at t=8 in terms of P, using continuous compounding.
(b) Use your expression to solve for the initial deposit P. (You may leave your answer in exact form involving exponentials and natural logarithms.)

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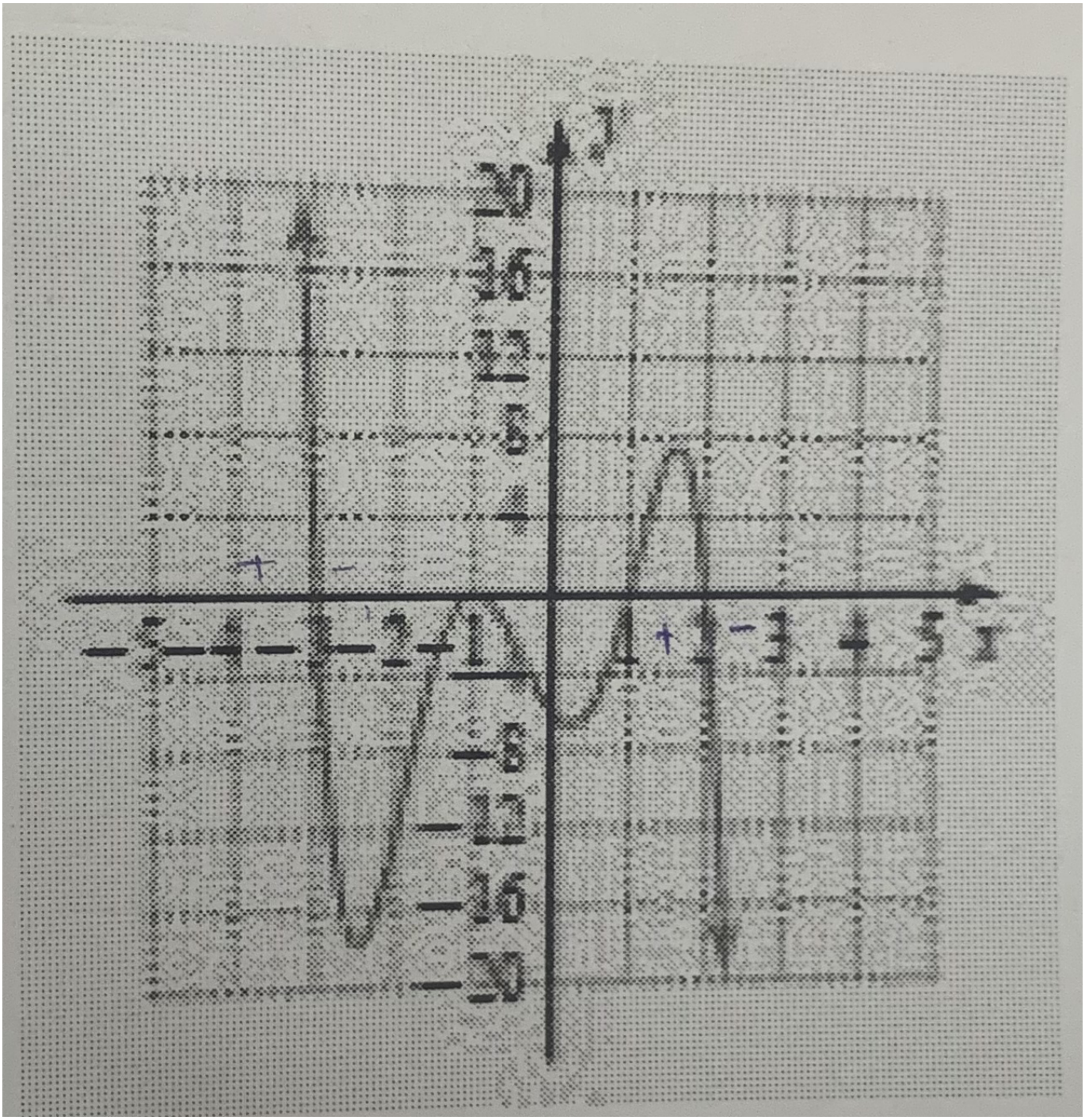
7. (40 points)
a) Solve the following inequality and draw the solution sets on the real line: $\frac{1}{x+1} > \frac{2}{x-1}$

b) Solve the equation

$$\log_3(x+1) + \log_3(x+4) - 2 = 0$$

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8. (40 points) For the polynomial graph below (Assume all zeroes are real).
(a) state whether the degree of the function is even or odd;
(b) use the graph to name the zeroes of f, then state whether their multiplicity is even or odd;
(c) state the minimum possible degree of f and write one possible function for f in factored form;
(d) estimate the domain and range.



9. (30 points) The proportion P of new cars with a global positioning system (GPS) can be modelled by the following function:

$P(t) = \frac{0.8}{1+1.87e^{-0.16t}}$

Given that t=0 represent 2006, t = 1 represent 2007, and so on.

(a) What proportion of new cars in 2006 had a GPS?

(b) Determine the maximum proportion of new cars that have a GPS?

(c) When will 75% of new cars have a GPS?